CHAPTER 15

ENVIRONMENT AND SUSTAINABLE DEVELOPMENT

CHAPTER ESSENCE AT A GLANCE:
Preserve ecological equilibrium: carrying capacity vs absorptive capacity, global warming and ozone depletion, land degradation, and sustainable development strategies.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Four Functions of Environment:** Supplies renewable and non-renewable resources, assimilates waste generated by human activity, sustains life by providing genetic biodiversity, and provides aesthetic scenery.

• **Carrying Capacity Threshold Invariant:** Environmental crisis occurs when resource extraction exceeds resource regeneration rates, OR when waste generation exceeds the absorptive capacity of the ecosystem.

• **Global Environmental Challenges:** Global Warming (greenhouse gases like CO_2 raising global temperatures and melting polar ice caps) and Ozone Depletion (CFCs destroying the stratospheric ozone layer, exposing Earth to harmful UV rays).

• **Sustainable Development Strategies:** Meeting present needs without compromising the ability of future generations to meet their own needs: shifting to solar or wind renewable energy, CNG in transport, mini-hydel plants, traditional bio-pest control, and organic composting.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$\text{Carrying Capacity Condition} \iff \text{Extraction Rate} \leq \text{Regeneration Rate} \quad \text{AND} \quad \text{Waste} \leq \text{Absorption}$$

Ecological Dynamic Equilibrium Invariant

$$\text{Sustainable Development} \iff \text{Present Human Well-being} + \text{Preserved Capital for Future Generations}$$

Brundtland Intergenerational Equity Model

$$\text{Environmental Crisis Trigger} \iff \text{Waste Discharge Rate} > \text{Ecological Assimilative Capacity}$$

Environmental Degradation Disequilibrium

$$\text{Clean Energy Transition} = \text{Solar Photovoltaic} + \text{Wind Turbines} + \text{Mini-Hydel} + \text{Biomass}$$

Renewable Decarbonization Portfolio

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Absorptive capacity is the ability of the environment to absorb and degrade waste without creating environmental damage; exceeding it leads to pollution crises.

• **Alert:** Ozone depletion is caused by chlorofluorocarbons (CFCs) used in refrigerators and air conditioners, NOT by carbon dioxide; carbon dioxide causes global warming.

• **Alert:** The Brundtland Commission Report (1987) officially defined Sustainable Development as meeting present needs without compromising future generations.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define Sustainable Development according to the Brundtland Commission. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs. [1 Mark]

• **Core Principle:** Emphasizes intergenerational equity and the conservation of natural capital. [1 Mark]

2. Distinguish between Carrying Capacity and Absorptive Capacity of the environment. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Carrying Capacity:** The condition where resource extraction does not exceed resource regeneration rates, keeping resource stocks intact. [1 Mark]

• **Absorptive Capacity:** The ability of the environment to absorb, assimilate, and degrade waste products without environmental breakdown. [1 Mark]

3. Explain the four primary functions performed by the Environment. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Supplies Resources:** Provides renewable resources (trees, water) and non-renewable minerals (coal, petroleum) for production. [0.75 Mark]

• **Assimilates Waste:** Naturally decomposes and absorbs waste matter generated by human consumption and industrial processes. [0.75 Mark]

• **Sustains Life:** Maintains vital biodiversity, oxygen atmosphere, and water cycles supporting biological life. [0.75 Mark]

• **Provides Aesthetic Services:** Offers natural scenic beauty, forests, mountains, and oceans improving human quality of life. [0.75 Mark]

4. Distinguish between Global Warming and Ozone Depletion. State the primary chemical culprit responsible for each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Global Warming:** The gradual increase in the average temperature of Earth's atmosphere caused by the buildup of greenhouse gases like Carbon Dioxide (CO_2) and methane. [1.5 Marks]

• **Ozone Depletion:** The thinning of the protective stratospheric ozone layer caused by Chlorofluorocarbons (CFCs), exposing Earth to harmful ultraviolet (UV) radiation. [1.5 Marks]

5. Enumerate any three major causes of Land Degradation in India. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Deforestation and Unsustainable Fuelwood Extraction:** Clearing forests for agriculture and logging strips topsoil of binding vegetation cover. [1 Mark]

• **Overgrazing:** Excessive cattle grazing destroys natural grass cover, leading to severe wind and water soil erosion. [1 Mark]

• **Over-irrigation and Chemical Overuse:** Excessive canal watering causes waterlogging and soil salinity; chemical fertilizers alter soil chemical composition. [1 Mark]

6. State any three strategies that can be adopted to achieve Sustainable Development in India. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Use of Non-Conventional Energy:** Shifting from thermal coal power to solar photovoltaic cells, wind turbines, and mini-hydel projects. [1 Mark]

• **Promoting Clean Urban Fuels:** Replacing diesel and petrol with Compressed Natural Gas (CNG) in public transport fleets to curb air pollution. [1 Mark]

• **Bio-Pest Control and Organic Manures:** Adopting traditional neem-based bio-pesticides and organic composting to protect soil biodiversity. [1 Mark]

7. Explain why the world is facing an acute Environmental Crisis today. Contrast the pre-industrial period with the modern industrial period. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Pre-Industrial Balance:** In earlier times, demand for environmental resources was well within carrying capacity; extraction was slower than natural regeneration, and waste was easily assimilated. **[2 Marks]**

• **Modern Industrial Reversal:** Massive population explosion and industrial consumerism have pushed resource extraction far beyond regeneration rates; toxic chemical wastes exceed absorptive capacity, causing air and water pollution, species extinction, and climate crises. **[2 Marks]**